Java2: Lesson 6 – Lab Project

1. A **summary** at the top, **documentation** at the bottom and **comments** where necessary.
   - Recordings and interviews are an exception, but they should be included in your submission code.
2. **Cite** the source of your information.
3. Don't put anything in your code if you don't **understand 100%**.

1

70 points

5-min challenge: **you will get full credit after you reproduce your work in class/interview.**

Write a GUI program to take an integer from a user and determine whether it is an even or odd number.

a. Install ***Scene Builder*** to design your GUI window.
   https://www.jetbrains.com/help/idea/opening-fxml-files-in-javafx-scene-builder.html
   https://openjfx.io/openjfx-docs/
   https://bit.ly/3SCFNhU

b. Submit **FXML** file (.fxml), **Controller** class (.java) and main **application** class (.java)

Note:
*Explain the code you are writing.
*Perform **two consecutive recordings** for each task.

*Follow the **standard routine** for each interview.
*1. **Share the entire screen** (https://bit.ly/3lNpkJV)*
*2. Use only one screen to share (**Win + P** > **PC screen only**)*
*3. Display the running programs (**Alt + TAB**)*
4. Get your **IDE (IntelliJ)** ready
5. **Presentation mode** (**Ctrl + ~`**)

** **Editing** your recordings is considered **cheating** and will not be tolerated.*

File(s) to submit	J206_1.java, *controller.java, *.fxml J206_1.png, YouTube link

2

130 points

Modify each code and mark **//*** where you modified.

a. This "how-to" assignment has no plagiarism issue (except **2-11** ~ **2-13**) as long as you submit your own work files.
b. Submit only **one** .java file for each problem.
c. Use copy & paste method.
d. Submit screenshots of each window.
e. Practice enough so that you can create any required component quickly and efficiently.

2-1. https://liveexample.pearsoncmg.com/html/LoanCalculator.html
https://liveexample.pearsoncmg.com/html/Loan.html
Take **three numbers** and calculate the **sum** and the **multiplication** of the numbers.

2-2. https://liveexample.pearsoncmg.com/html/HandleEvent.html
Display a welcome message when the 'OK' button is clicked and remove it when 'Cancel' is clicked."

2-3. https://liveexample.pearsoncmg.com/html/ControlCircle.html
Color the circle with your favorite color.

	2-4. https://liveexample.pearsoncmg.com/html/AnonymousHandlerDemo.html Include diagonal move buttons and adjust the step size for all movements. 2-5. https://liveexample.pearsoncmg.com/html/MouseEventDemo.html Modify both the text color and its content. 2-6. https://liveexample.pearsoncmg.com/html/ControlCircleWithMouseAndKey.html https://liveexample.pearsoncmg.com/html/ControlCircle.html Transform the circle into a square. 2-7. https://liveexample.pearsoncmg.com/html/FlagRisingAnimation.html Select any image you prefer (http://liveexample.pearsoncmg.com/book/image/us.gif) Adjust both the speed and the direction of its movement (e.g., from left to right). 2-8. https://liveexample.pearsoncmg.com/html/FadeTransitionDemo.html Allow the animation to **play** and **pause** upon clicking the play and pause **buttons**. 2-9. https://liveexample.pearsoncmg.com/html/TimelineDemo.html Adjust the text color, font, message content, and display rate. 2-10. https://liveexample.pearsoncmg.com/html/BounceBallControl.html https://liveexample.pearsoncmg.com/html/BallPane.html Change the color and the speed of the ball. 2-11*. https://liveexample.pearsoncmg.com/html/KeyEventDemo.html Display special keys like 'Shift,' 'Ctrl,' and 'Alt' (or 'Shift,' 'Ctrl,' 'Option,' 'Command' for Mac) along with regular keys when pressed. 2-12*. https://liveexample.pearsoncmg.com/html/DisplayResizableClock.html https://liveexample.pearsoncmg.com/html/ClockAnimation.html https://liveexample.pearsoncmg.com/html/ClockPane.html Adjust the thickness of the hour and minute hands. Systematically add more numbers around the clock and relocate the 'digital watch'. 2-13*. https://liveexample.pearsoncmg.com/html/PathTransitionDemo.html Make the rectangle trace the perimeter of a square.
	File name: J206_21,22, ~ 210, ~ , 213.java J206_21,22, ~ 210, ~ , 213.png

***Advanced students: Address the '*TicTacToe*' question in the 'challenge problem' folder for extra credits.**